On the Non-Elliptic Constraint Structure of General Relativity

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The non-elliptic principal-symbol structure of the Einstein constraint equations admits a realization within a purely Riemannian compatibility framework with internal normal gauge only. A triple $(h, K, \nabla^{\perp})$ determines a canonical metric connection on $TM \oplus N$ whose flatness is equivalent to the Gauss–Codazzi–Ricci system and defines a first-order internal-gauge constraint geometry on an affine space of orthogonal connections. In codimension one, the two Einstein constraint principal-symbol classes coincide with canonical contractions of the compatibility curvature decomposition.

At flat compatible backgrounds with $K_0 = 0$ and for $n \geq 2$, the reduced principal symbol fails injectivity modulo internal normal gauge, and no admissible internal gauge condition restores ellipticity. In frozen flat charts, the mixed sector admits a local Hessian potential description and generates an infinite-dimensional reduced formal tangent of the flat locus.

I. INTRODUCTION

a. Standing background. Fix a smooth compact boundaryless manifold M, a Riemannian metric h on TM, a Euclidean vector bundle $(N, g_N) \to M$, and the Levi–Civita connection ∇^h of h. The background is fixed globally: h is not varied.

b. Gauge convention. Gauge quotient uses internal normal gauge $\Gamma(\mathrm{SO}(N))$ only; diffeomorphisms are not quotiented.

c. Purpose and scope. The Einstein constraint equations are non-elliptic at principal-symbol level[7]. The same degeneracy appears in the intrinsic compatibility-flatness system of the canonical orthogonal connection. The degeneracy survives reduction by admissible internal gauge fixing. In codimension one, the corresponding momentum and Hamiltonian principal-symbol classes coincide with contractions of the mixed and Gauss blocks at flat compatible backgrounds. The principal-symbol analysis is formulated invariantly as a bundle morphism over the cosphere bundle S^*M, and the paper gives a structural microlocal classification of a first-order gauge-natural operator.

A. Main Results

The principal structural theorems are:

(i) codimension-one momentum-symbol reduction and universality (Theorem VI.2 and Corollary VI.3),

(ii) codimension-one Hamiltonian-symbol reduction from Gauss scalar contraction (Theorem VI.6 and Corollary VI.7),

(iii) reduced non-ellipticity modulo internal gauge at flat compatible backgrounds with $K_0 = 0$ and impossibility of admissible elliptic restoration (Theorems IV.14 and IV.16),

(iv) local potential parametrization of frozen mixed kernel and infinite-dimensional reduced tangent (Theorem V.1 and Theorem V.6).

At flat compatible backgrounds with $K_0 = 0$ and for $n \geq 2$, the reduced principal symbol of the compatibility flatness system fails injectivity at every nonzero covector, and no admissible internal gauge fixing restores ellipticity. The mixed-sector kernel has dimension k at each such covector (Theorem IV.7).

II. CANONICAL CONNECTION AND GCR BLOCKS

A. Notation and Conventions

Use the standing background fixed above. The bundle $(N, g_N) \to M$ has rank k and metric connection $\nabla^\perp$. Assume M is oriented henceforth, and let $*$ denote the Hodge star induced by h.

Fix

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z, \tag{1}$$

for any connection ∇ on a vector bundle.

Set

$$E := TM \oplus N, \qquad g_E := h \oplus g_N. \tag{2}$$

A GCR triple is a datum

$$(h, K, \nabla^\perp), \qquad K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N). \tag{3}$$

For each $X \in TM$, define

$$K_X : TM \to N, \qquad K_X(Y) := K(X,Y), \tag{4}$$

and its $h \oplus g_N$-adjoint

$$K_X^* : N \to TM, \qquad h(K_X^* \nu, Y) = \langle \nu, K_X Y \rangle_N. \tag{5}$$

Equivalently, regard

$$K \in \Omega^1(M, \mathrm{Hom}(TM, N)), \qquad K(X) := K_X, \tag{6}$$

and

$$K^* \in \Omega^1(M, \mathrm{Hom}(N, TM)), \qquad K^*(X) := K_X^*. \tag{7}$$

Define also

$$A \in \Omega^1(M, \mathrm{Hom}(N, TM)), \qquad A(X) := K_X^*. \tag{8}$$

Equivalently, $A_\nu X := K_X^* \nu$.

All bundle-valued wedges are composition wedges:

$$(\alpha \wedge \beta)(X,Y) = \alpha_X \circ \beta_Y - \alpha_Y \circ \beta_X. \tag{9}$$

For homogeneous form degrees p, q, the graded commutator is

$$[\alpha, \beta]_{\mathrm{gr}} := \alpha \wedge \beta - (-1)^{pq} \beta \wedge \alpha, \qquad \alpha \in \Omega^p(M, \mathrm{End}(E)),\ \beta \in \Omega^q(M, \mathrm{End}(E)). \tag{10}$$

Define

$$(K^* \wedge K)(X,Y) := K_X^* \circ K_Y - K_Y^* \circ K_X \in \mathrm{End}(TM), \tag{11}$$

$$(K \wedge K^*)(X,Y) := K_X \circ K_Y^* - K_Y \circ K_X^* \in \mathrm{End}(N). \tag{12}$$

B. Notation for induced covariant derivatives

Let D^T denote the exterior covariant derivative on $\mathrm{End}(TM)$ induced by ∇^h. Let $D^\perp$ denote the exterior covariant derivative on $\mathrm{End}(N)$ induced by $\nabla^\perp$. Let

$$\nabla_X^{\mathrm{Hom}} T := \nabla_X^\perp \circ T - T \circ \nabla_X^h, \tag{13}$$

for $T \in \Gamma(\mathrm{Hom}(TM, N))$, and let D^{Hom} be its exterior covariant derivative on $\mathrm{Hom}(TM, N)$-valued forms.

a. Type summary. The curvature blocks satisfy

$$\mathscr{G} \in \Omega^2(M, \mathrm{End}(TM)), \quad \mathscr{C} \in \Omega^2(M, \mathrm{Hom}(TM, N)), \quad \mathscr{S} \in \Omega^2(M, \mathrm{End}(N)).$$

Theorem II.1 (Canonical Assembly)**.** *Given a GCR triple $(h, K, \nabla^\perp)$, there exists a unique metric connection $\nabla^\oplus$ on $E = TM \oplus N$ such that for every $X \in TM$:*

(i) the $TM \to TM$ block of $(\nabla^\oplus)_X$ equals ∇^h_X,

(ii) the $N \to N$ block of $(\nabla^\oplus)_X$ equals $\nabla^\perp_X$,

(iii) the $TM \to N$ block of $(\nabla^\oplus)_X$ equals K_X.

Equivalently, the $N \to TM$ block is $-K^*_X$.

Proof. Define

$$\nabla^\oplus_X (Y \oplus \nu) := (\nabla^h_X Y - K^*_X \nu) \oplus (\nabla^\perp_X \nu + K_X Y). \tag{14}$$

Metricity follows from adjointness of K_X and K^*_X:

$$\begin{aligned} X\langle Y_1 \oplus \nu_1, Y_2 \oplus \nu_2 \rangle_E = \langle \nabla^h_X Y_1, Y_2 \rangle_h + \langle Y_1, \nabla^h_X Y_2 \rangle_h \\ + \langle \nabla^\perp_X \nu_1, \nu_2 \rangle_N + \langle \nu_1, \nabla^\perp_X \nu_2 \rangle_N, \end{aligned}$$

and the mixed terms cancel:

$$-\langle K^*_X \nu_1, Y_2 \rangle_h + \langle K_X Y_1, \nu_2 \rangle_N - \langle Y_1, K^*_X \nu_2 \rangle_h + \langle \nu_1, K_X Y_2 \rangle_N = 0.$$

For uniqueness, let $\widetilde{\nabla}$ satisfy (i)–(iii). Write

$$\widetilde{\nabla} = \nabla^h \oplus \nabla^\perp + \Phi, \tag{15}$$

with $\Phi_X \in \mathrm{End}(E)$. Conditions (i)–(iii) force the $TM \to TM$ and $N \to N$ blocks of Φ_X to vanish and the $TM \to N$ block to be K_X. Metricity forces Φ_X skew-adjoint, hence the $N \to TM$ block is $-K^*_X$. Therefore $\widetilde{\nabla} = \nabla^\oplus$. □

Define the block perturbation 1-form

$$\Psi_X := \begin{pmatrix} 0 & -K^*_X \\ K_X & 0 \end{pmatrix}, \qquad \nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi. \tag{16}$$

C. Curvature Decomposition

Define operator curvature on E by

$$R^\oplus(X, Y) := [\nabla^\oplus_X, \nabla^\oplus_Y] - \nabla^\oplus_{[X,Y]} \in \mathrm{End}(E), \tag{17}$$

and define

$$\mathscr{F}^\oplus(X, Y) := R^\oplus(X, Y) \in \Omega^2(M, \mathfrak{so}(E)). \tag{18}$$

In any adapted orthonormal frame, with connection matrix Ω of $\nabla^\oplus$,

$$\mathscr{F}^\oplus = d\Omega + \Omega \wedge \Omega. \tag{19}$$

Proposition II.2 (Explicit curvature blocks)**.** *The curvature decomposition is functorial with respect to the orthogonal splitting $E = TM \oplus N$. With respect to $E = TM \oplus N$, let $\mathscr{C}^* \in \Omega^2(M, \mathrm{Hom}(N, TM))$ denote the fiberwise adjoint of $\mathscr{C}$ with respect to h and $\langle \cdot, \cdot \rangle_N$.*

$$\mathscr{F}^{\oplus} = \begin{pmatrix} \mathscr{G} & -\mathscr{C}^* \\ \mathscr{C} & \mathscr{S} \end{pmatrix}, \tag{20}$$

where

$$\mathscr{G} = R^h - (K^* \wedge K) = R^h - (A \wedge K), \tag{21}$$

$$\mathscr{C} = D^{\mathrm{Hom}} K, \tag{22}$$

$$\mathscr{S} = R^{\perp} - (K \wedge K^*) = R^{\perp} - (K \wedge A). \tag{23}$$

Proof. From $\nabla^{\oplus} = \nabla^{\mathrm{prod}} + \Psi$, with $\nabla^{\mathrm{prod}} := \nabla^h \oplus \nabla^{\perp}$,

$$\mathscr{F}^{\oplus} = \mathscr{F}_{\mathrm{prod}} + D^{\mathrm{prod}} \Psi + \Psi \wedge \Psi. \tag{24}$$

The $TM \to N$ block of $\Psi \wedge \Psi$ vanishes because compositions of two off-diagonal maps are diagonal. The $TM \to N$ block of $D^{\mathrm{prod}} \Psi$ is exactly $D^{\mathrm{Hom}} K$, which gives (22).

For $Z \in TM$,

$$\begin{aligned} ((\Psi \wedge \Psi)_{TT}(X,Y))Z &= \Psi_X(\Psi_Y Z) - \Psi_Y(\Psi_X Z) \\ &= -K_X^*(K_Y Z) + K_Y^*(K_X Z) \\ &= -(K^* \wedge K)(X,Y)Z, \end{aligned}$$

yielding (21). For $\nu \in N$,

$$\begin{aligned} ((\Psi \wedge \Psi)_{NN}(X,Y))\nu &= \Psi_X(\Psi_Y \nu) - \Psi_Y(\Psi_X \nu) \\ &= -K_X(K_Y^* \nu) + K_Y(K_X^* \nu) \\ &= -(K \wedge K^*)(X,Y)\nu, \end{aligned}$$

yielding (23). □

Theorem II.3 (Integrability Equivalence)**.** *For a GCR triple $(h, K, \nabla^{\perp})$, the following are equivalent:*

(i) $\mathscr{F}^{\oplus} = 0$*;*

(ii) the three equations (21)–(23) *vanish, equivalently the Gauss–Codazzi–Ricci system.*

Equivalently, flatness of the assembled metric connection is equivalent to vanishing of the three orthogonal curvature blocks.

Proof. By Proposition II.2, the blocks of $\mathscr{F}^{\oplus}$ are exactly $(\mathscr{G}, \mathscr{C}, \mathscr{S})$, so $\mathscr{F}^{\oplus} = 0$ iff each block vanishes. In an adapted orthonormal frame these recover the classical component Gauss/Codazzi/Ricci identities. □

Proposition II.4 (Internal gauge equivariance)**.** *The assignment*

$$(h, K, \nabla^{\perp}) \longmapsto \nabla^{\oplus}$$

is equivariant under internal normal gauge $\Gamma(\mathrm{SO}(N))$. Consequently, $\mathscr{F}^{\oplus}$ and its blocks $(\mathscr{G}, \mathscr{C}, \mathscr{S})$ transform covariantly.

Proof. The gauge action is blockwise conjugation on $\nabla^{\perp}$ and tensor action on K. The assembled form $\nabla^{\oplus} = \nabla^h \oplus \nabla^{\perp} + \Psi(K)$ is preserved under this action. Curvature transforms by conjugation, and block projections are equivariant. □

D. The Bianchi Identity

Fix canonical pairing

$$\langle A,B\rangle_{\text{can}} := -\text{tr}(AB), \qquad A,B \in \mathfrak{so}(E). \tag{25}$$

For $\mathfrak{so}(E)$-valued forms, use

$$\langle \alpha,\beta\rangle := -\text{tr}(\alpha\wedge *\beta), \qquad \|\mathscr{F}^{\oplus}\|^2 := -\text{tr}(\mathscr{F}^{\oplus}\wedge *\mathscr{F}^{\oplus}). \tag{26}$$

This is the pointwise norm induced by $\langle\cdot,\cdot\rangle_{\text{can}}$ and h. The splitting

$$\mathfrak{so}(E) = \mathfrak{so}(TM) \oplus \big(\text{Hom}(TM,N)\oplus \text{Hom}(N,TM)\big) \oplus \mathfrak{so}(N)$$

is orthogonal under $-\text{tr}$, so cross terms vanish. Then the block-orthogonal split gives

$$\|\mathscr{F}^{\oplus}\|^2 = \|\mathscr{G}\|^2 + 2\|\mathscr{C}\|^2 + \|\mathscr{S}\|^2. \tag{27}$$

Theorem II.5 (Coupled Bianchi System)**.** *The identity $D^{\oplus}\mathscr{F}^{\oplus} = 0$ decomposes as*

$$D^T\mathscr{G} + [K^*,\mathscr{C}] = 0, \tag{28}$$

$$D^{\text{Hom}}\mathscr{C} + K\wedge\mathscr{S} - \mathscr{G}\wedge K = 0, \tag{29}$$

$$D^{\perp}\mathscr{S} + [K,\mathscr{C}^*] = 0. \tag{30}$$

Proof. Write

$$\Psi = \begin{pmatrix} 0 & -K^* \\ K & 0 \end{pmatrix}, \qquad \mathscr{F}^{\oplus} = \begin{pmatrix} \mathscr{G} & -\mathscr{C}^* \\ \mathscr{C} & \mathscr{S} \end{pmatrix}. \tag{31}$$

Using $D^{\oplus} = D^{\text{prod}} + [\Psi,\cdot]_{\text{gr}}$,

$$0 = D^{\oplus}\mathscr{F}^{\oplus} = D^{\text{prod}}\mathscr{F}^{\oplus} + [\Psi,\mathscr{F}^{\oplus}]_{\text{gr}}. \tag{32}$$

Since $\deg\Psi = 1$ and $\deg\mathscr{F}^{\oplus} = 2$, $[\Psi,\mathscr{F}^{\oplus}]_{\text{gr}} = \Psi\wedge\mathscr{F}^{\oplus} - \mathscr{F}^{\oplus}\wedge\Psi$. Block projection of the identity gives: the (TT) block yields (28), the (TN) block yields (29), and the (NN) block yields (30)[3]. □

Remark II.6 (Classical immersion consequence)**.** If $U\subset M$ is simply connected and $\mathscr{F}^{\oplus}|_U = 0$, the fundamental theorem of submanifolds yields a local immersion with prescribed $(K,\nabla^{\perp})$, unique up to rigid motion and parallel normal gauge[2] (Vol. II, Ch. VII, §4).

E. Symbol Calculus for Connection-Valued Operators

Lemma II.7 (Symbol of covariant exterior derivative)**.** *Let ∇ be a connection on a vector bundle $V\to M$. For $\alpha\in\Omega^p(M,V)$ and $\xi\in T_x^*M\setminus\{0\}$,*

$$\sigma_\xi(D^{\nabla})(\alpha) = \xi\wedge\alpha.$$

Proof. In a local trivialization, $D^{\nabla} = d + A$ with A order zero, so the principal symbol is that of d, namely wedge with ξ. □

Lemma II.8 (Linearization of curvature and its symbol)**.** *For a connection ∇ on V, the curvature map $\nabla\mapsto F^{\nabla}$ has linearization*

$$DF^{\nabla}(\eta) = D^{\nabla}\eta, \qquad \eta\in\Omega^1(M,\text{End}(V)).$$

Hence

$$\sigma_\xi(DF^{\nabla})(\eta) = \xi\wedge\eta.$$

Proof. The first identity is the standard curvature variation formula. The symbol identity follows from Lemma II.7. □

Lemma II.9 (Gauge infinitesimal action by slots)**.** *In the block decomposition $E = TM \oplus N$ and internal gauge $\phi \in \Gamma(\mathfrak{so}(N))$, the infinitesimal action satisfies*

$$\delta_\phi(\nabla^\perp) = D^\perp \phi, \qquad \delta_\phi K = -\phi \cdot K.$$

Thus the action is order one on the normal-connection slot and order zero on the mixed slot; the mixed component is algebraic and linear in the background K.

Proof. Differentiate the gauge action $\nabla^\perp \mapsto g^{-1}\nabla^\perp g$ and $K \mapsto g^{-1} \cdot K$ at $g = \exp(t\phi)$. □

III. CONSTRAINT GEOMETRY OF FLATNESS

Fix $E = TM \oplus N$ with orthogonal bundle metric $g_E = h \oplus g_N$ and fixed splitting type.

Definition III.1 (Connection configuration space)**.** Define

$$\mathscr{A}_\oplus := \{\nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi(K) : \ \nabla^\perp \in \Omega^1(M, \mathfrak{so}(N)), \ K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)\}.$$

Definition III.2 (Internal gauge group and infinitesimal action)**.** Let

$$\mathscr{G} := \Gamma(\mathrm{SO}(N)).$$

Its action is

$$\nabla^\perp \mapsto g^{-1}\nabla^\perp g, \qquad K \mapsto g^{-1} \cdot K.$$

For $\phi \in \Gamma(\mathfrak{so}(N))$, define

$$\tilde{\phi} = \begin{pmatrix} 0 & 0 \\ 0 & \phi \end{pmatrix} \in \Gamma(\mathfrak{so}(E)).$$

Infinitesimally,

$$\delta_\phi \nabla^\oplus = D^{\nabla^\oplus} \tilde{\phi}.$$

With the convention $K \mapsto g^{-1} \cdot K$,

$$\delta_\phi(\nabla^\perp) = D^\perp \phi, \qquad \delta_\phi K = -\phi \cdot K.$$

Proposition III.3. *The $\Gamma(\mathrm{SO}(N))$-action preserves $\mathscr{A}_\oplus$.*

Proof. Orthogonal automorphisms preserve metricity, block type, and the symmetry condition on K. □

Definition III.4 (Constraint operator and constraint locus)**.** Define the gauge-equivariant first-order operator

$$\Phi : \mathscr{A}_\oplus \to \Omega^2(M, \mathfrak{so}(E)), \qquad \Phi(\nabla) := F^\nabla,$$

so $\Phi(\nabla^\oplus) = \mathscr{F}^\oplus$ for the assembled connection $\nabla^\oplus$, and define the constraint locus

$$\mathscr{Z}_{\mathrm{flat}} := \Phi^{-1}(0) \subset \mathscr{A}_\oplus.$$

Definition III.5 (First-order internal gauge constraint system)**.** An internal gauge first-order constraint system on a connection space is a gauge-equivariant first-order differential operator whose zero set is gauge-invariant.

Remark III.6. Orthogonal compatibility is the system $(\mathscr{A}_\oplus, \Phi, \mathscr{G})$ with constraint locus $\mathscr{Z}_{\text{flat}}$.

The triple $(\mathscr{A}_\oplus, \Phi, \mathscr{G})$ is a first-order gauge-natural constraint operator system. Its linearization is covariant exterior differentiation (Theorem III.8), and its principal symbol is wedge with the covector by Lemma II.8.

Lemma III.7 (Gauge equivariance of Φ)**.** *The curvature map $\Phi : \mathscr{A}_\oplus \to \Omega^2(M, \mathfrak{so}(E))$ is $\mathscr{G}$-equivariant:*

$$\Phi(g^{-1} \cdot \nabla) = g^{-1}\Phi(\nabla)g.$$

Proof. For gauge transform $\nabla \mapsto g^{-1}\nabla g$, standard curvature covariance gives

$$F^{g^{-1}\nabla g} = g^{-1}F^{\nabla}g.$$

□

Theorem III.8 (Linearization of the constraint operator)**.** *The map Φ is first-order nonlinear, and for $\eta \in T_{\nabla^\oplus}\mathscr{A}_\oplus$,*

$$D\Phi|_{\nabla^\oplus}(\eta) = D^{\nabla^\oplus}\eta.$$

Proof. For a smooth variation $\nabla_t = \nabla^\oplus + t\eta$, the curvature variation identity gives

$$\frac{d}{dt}\Big|_{t=0} F^{\nabla_t} = D^{\nabla^\oplus}\eta.$$

Hence the Fréchet derivative is $D^{\nabla^\oplus}\eta$. □

a. Bridge to codimension-one symbols. At flat codimension-one compatible backgrounds, contractions of the mixed and Gauss blocks recover the momentum and Hamiltonian principal-symbol classes as equalities of principal-symbol bundle morphisms. Section VI gives the explicit identifications.

IV. SYMBOL AND REDUCED SYMBOL AT FLAT COMPATIBLE BACKGROUNDS

A. The Principal Symbol

Define

$$\mathscr{D}(h, K, \nabla^\perp) := (\mathscr{G}, \mathscr{C}, \mathscr{S}). \tag{33}$$

Let $\mathbf{L}_{(h,K,\nabla^\perp)} := D\mathscr{D}|_{(h,K,\nabla^\perp)}$. Set $\mathbf{L}_{\text{full}} := D\Phi|_{\nabla^\oplus}$ and define the order-zero bundle embedding

$$\iota : \Omega^1(M, \mathfrak{so}(N)) \oplus \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \hookrightarrow \Omega^1(M, \mathfrak{so}(E))$$

by

$$\iota(\delta\nabla^\perp, \delta K)_X = \begin{pmatrix} 0 & -(\delta K)_X^* \\ (\delta K)_X & (\delta\nabla^\perp)_X \end{pmatrix}.$$

Set $\eta := \iota(\delta\nabla^\perp, \delta K)$. Reduced ellipticity means injectivity of the principal symbol after quotienting the domain by the principal-symbol image of internal infinitesimal gauge directions.

Definition IV.1 (Flat point and frozen flat chart)**.** A *flat point* is a connection $\nabla^\oplus \in \mathscr{Z}_{\text{flat}}$, equivalently $F^{\nabla^\oplus} = 0$. A *frozen flat chart at a point x* is a choice of local coordinates and normal gauge such that the Levi–Civita coefficients of h and the normal-connection coefficients of $\nabla^\perp$ vanish at x (equivalently, the connection matrices vanish at x). On a simply connected open set U, *frozen flat on U* means these coefficients vanish identically on U. In symbol-level arguments using gauge invisibility in the mixed slot, we additionally impose the flat compatible background condition $K_0 = 0$.

Theorem IV.2 (Principal symbol of the curvature linearization)**.** *For any background $\nabla^{\oplus} \in \mathscr{A}_{\oplus}$ and $\eta \in T_{\nabla^{\oplus}}\mathscr{A}_{\oplus}$,*

$$\sigma_{\xi}(\mathbf{L}_{\text{full}})(\eta) = \xi \wedge \eta, \qquad \mathbf{L}_{\text{full}} = D\Phi|_{\nabla^{\oplus}}.$$

Proof. This is Lemma II.8 with $V = E$. □

Corollary IV.3 (Block projection of the principal symbol)**.** *Let $\eta = \iota(\delta\nabla^{\perp}, \delta K)$. Then*

$$\mathrm{pr}_{TT,TN,NN}\big(D\Phi|_{\nabla^{\oplus}}(\eta)\big) = D\mathscr{D}|_{(h,K,\nabla^{\perp})}(\delta\nabla^{\perp}, \delta K),$$

$$\mathrm{pr}_{TT,TN,NN}\big(\sigma_{\xi}(\mathbf{L}_{\text{full}})(\eta)\big) = \mathrm{pr}_{TT,TN,NN}(\xi \wedge \eta).$$

Proof. The first identity is the block definition of $\mathscr{D}$. The second follows from Theorem IV.2. □

Lemma IV.4 (Mixed-slot gauge linearization at $K_0 = 0$)**.** *For internal gauge parameter $\phi \in \Gamma(\mathfrak{so}(N))$ and background $(h, K_0, \nabla^{\perp})$,*

$$\delta_{\phi} K = -\,\phi \cdot K_0.$$

Hence at backgrounds with $K_0 = 0$, the full linearized gauge action on the mixed slot vanishes:

$$\mathrm{pr}_{\delta K}\big(D(\text{gauge})|_{\nabla^{\oplus}}\phi\big) = 0.$$

$$\mathrm{pr}_{\delta K}\big(\sigma_{\xi}(D(\text{gauge})|_{\nabla^{\oplus}})\phi\big) = 0.$$

Proof. The first identity is Lemma II.9. It is algebraic in ϕ and linear in K_0, so both the full mixed component and its principal symbol vanish at $K_0 = 0$. □

At a flat compatible background with $K_0 = 0$ and with $\delta\nabla^{\perp} = 0$, the mixed block reduces to

$$L(\delta K) := D^{\text{Hom}}(\delta K), \tag{34}$$

with symbol

$$(\sigma_{\xi}\,\delta K)_{ij}{}^{\alpha}{}_{k} = \xi_i\,\delta K^{\alpha}_{jk} - \xi_j\,\delta K^{\alpha}_{ik}. \tag{35}$$

Remark IV.5 (Coordinate model check)**.** The component symbol display is a local verification of the invariant principal-symbol formula from Theorem IV.2.

Remark IV.6 (Microlocal notation)**.** Let $\pi : S^*M \to M$ be the cosphere bundle projection, and write pullback bundles as $\pi^*(\cdot)$.

Theorem IV.7 (Intrinsic Non-Ellipticity)**.** *Let $n \geq 2$. For every nonzero $\xi \in T^*_x M$,*

$$\sigma_{\xi}(L) : \mathrm{Sym}^2(T^*_x M) \otimes N_x \longrightarrow \Lambda^2(T^*_x M) \otimes T^*_x M \otimes N_x \tag{36}$$

has

$$\ker \sigma_{\xi}(L) = \{\xi \odot \xi \otimes v : v \in N_x\}. \tag{37}$$

Hence $\dim\ker\sigma_{\xi}(L) = k$. *At a flat compatible background with $K_0 = 0$ in a frozen flat chart, infinitesimal internal gauge action does not contribute to the principal symbol in the mixed δK slot, so these kernel directions survive principal gauge reduction.*

Proof. The equation is $\xi \wedge \delta K = 0$, equivalently $\delta K \in \ker(\xi \wedge \cdot)$. For $\xi \neq 0$, the kernel of

$$\xi \wedge \cdot : \mathrm{Sym}^2(T^*_x M) \otimes N_x \to \Lambda^2(T^*_x M) \otimes T^*_x M \otimes N_x$$

consists exactly of tensors of the form $\xi \odot \xi \otimes v$ with $v \in N_x$. Hence

$$\ker \sigma_{\xi}(L) = \{\xi \odot \xi \otimes v : v \in N_x\}.$$

For reduced-symbol persistence at $K_0 = 0$, infinitesimal internal gauge action on K is

$$\delta_{\phi} K = -\,\phi \cdot K, \tag{38}$$

which is linear in K and vanishes at $K_0 = 0$ in the mixed slot. □

B. Analytic Consequences

Proposition IV.8 (Frozen Codazzi non-ellipticity)**.** *For $n \geq 2$, the frozen operator*

$$L = D^{\mathrm{Hom}} : \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \to \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes N)$$

is non-elliptic.

Proof. Theorem IV.7 gives $\dim\ker\sigma_\xi(L) = k$ for every nonzero ξ. Therefore $\sigma_\xi(L)$ is never injective for $\xi \neq 0$, so ellipticity fails. □

Theorem IV.9 (Non-Fredholm Codazzi Linearization)**.** *Let M^n be compact, smooth, and without boundary, with $n \geq 2$. For every $s \in \mathbb{R}$, the frozen Codazzi operator*

$$L : H^s(\mathrm{Sym}^2 T^*M \otimes N) \to H^{s-1}(\Lambda^2 T^*M \otimes T^*M \otimes N)$$

is not Fredholm.

Proof. Fix $(x,\xi) \in S^*M$ and choose a nonzero amplitude a in $S_{(x,\xi)} = \ker\sigma_\xi(L)$. In local coordinates, take

$$u_\lambda(y) = e^{i\lambda\varphi(y)} a(y), \qquad d\varphi_x = \xi.$$

Then the principal term vanishes since $a(x) \in \ker\sigma_\xi(L)$, and

$$\|Lu_\lambda\|_{H^{s-1}} = O(\lambda^{-1})\|u_\lambda\|_{H^s}$$

after normalization $\|u_\lambda\|_{H^s} = 1$, this is a high-frequency approximate null sequence and rules out Fredholmness. This matches the classical criterion that ellipticity is equivalent to existence of a pseudodifferential parametrix and hence to Fredholmness on closed manifolds[4–6] (Vol. III, Ch. 20). Since Proposition IV.8 gives symbol non-ellipticity, L is not Fredholm. □

C. The Reduced Symbol

Throughout this subsection, work at a flat compatible background with $K_0 = 0$ in a frozen flat chart unless explicitly stated otherwise.

Define the frozen infinitesimal data bundle

$$\mathscr{U} := (\mathrm{Sym}^2 T^*M \otimes N) \oplus (T^*M \otimes \mathfrak{so}(N)),$$

the defect target bundle

$$\mathscr{V} := (\Lambda^2 T^*M \otimes \mathrm{End}(TM)) \oplus (\Lambda^2 T^*M \otimes T^*M \otimes N) \oplus (\Lambda^2 T^*M \otimes \mathfrak{so}(N)),$$

and the gauge-parameter bundle

$$\mathscr{P} := \mathfrak{so}(N).$$

At background $\nabla^\oplus \in \mathscr{A}_\oplus$, set

$$\mathbf{L}_{\mathrm{full}} := D\Phi|_{\nabla^\oplus} : \Gamma(\mathscr{U}) \to \Gamma(\mathscr{V}), \qquad \mathbf{G} := D(\text{gauge action})|_{\nabla^\oplus} : \Gamma(\mathscr{P}) \to \Gamma(\mathscr{U}).$$

For $(x,\xi) \in S^*M$, the reduced symbol bundle morphism is

$$\bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}}) : \mathscr{U}_x / \mathrm{im}\,\sigma_\xi(\mathbf{G}) \to \mathscr{V}_x.$$

Reduced ellipticity means fiberwise injectivity of $\bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}})$ for every $(x,\xi) \in S^*M$.

Definition IV.10 (Microlocal mixed kernel subbundle)**.** Define $S \subset \pi^*(\mathrm{Sym}^2 T^*M \otimes N)$ over S^*M by

$$S_{(x,\xi)} := \{\xi \odot \xi \otimes \nu : \nu \in N_x\}.$$

Proposition IV.11 (Microlocal kernel classification)**.** *For $n \geq 2$, S is a smooth rank-k $O(n)$-equivariant vector subbundle of $\pi^*(\mathrm{Sym}^2 T^*M \otimes N)$, and*

$$S_{(x,\xi)} = \ker \sigma_\xi(L) \subset \mathrm{Sym}^2(T_x^*M) \otimes N_x$$

for every $(x,\xi) \in S^*M$.

Proof. The fiberwise identity is Theorem IV.7. Smoothness and $O(n)$-equivariance follow from the tensorial formula $S_{(x,\xi)} = (\xi \odot \xi) \otimes N_x$. □

Lemma IV.12 (Gauge-image exclusion for frozen mixed slot)**.** *At flat compatible background with $K_0 = 0$, the subspace $S_{(x,\xi)}$ from Definition IV.10 has trivial intersection with $\mathrm{im}\,\sigma_\xi(\mathbf{G})$ in the mixed slot.*

Proof. By Lemma IV.4, the full mixed-slot gauge linearization vanishes at $K_0 = 0$, so its principal symbol projection to the mixed slot also vanishes. Hence no nonzero vector in $S_{(x,\xi)}$ lies in $\mathrm{im}\,\sigma_\xi(\mathbf{G})$ in that slot. □

Proposition IV.13 (Reduced symbol kernel contains the microlocal mixed subbundle)**.** *For every $(x,\xi) \in S^*M$ at a flat compatible background with $K_0 = 0$, the class of $S_{(x,\xi)} \oplus \{0\} \subset \mathscr{U}_x$ is contained in*

$$\ker \bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}}) \subset \mathscr{U}_x / \mathrm{im}\,\sigma_\xi(\mathbf{G}).$$

Proof. By Proposition IV.11 and Corollary IV.3, $S_{(x,\xi)} \oplus \{0\} \subset \ker \sigma_\xi(\mathbf{L}_{\mathrm{full}})$. By Lemma IV.12, these classes are nonzero in the quotient by $\mathrm{im}\,\sigma_\xi(\mathbf{G})$. □

Theorem IV.14 (Structural Non-Ellipticity Modulo Gauge)**.** *In the background-fixed internal-gauge setting with $\mathscr{G} = \Gamma(\mathrm{SO}(N))$, at each flat compatible background $\nabla^\oplus \in \mathscr{Z}_{\mathrm{flat}}$ with $K_0 = 0$ and for every nonzero covector ξ, the reduced principal symbol of $(\mathscr{A}_\oplus, \Phi, \mathscr{G})$ fails injectivity for $n \geq 2$.*

Proof. Embed the K-sector

$$U_K = \mathrm{Sym}^2(T_x^*M) \otimes N_x \hookrightarrow \mathscr{U}_x, \qquad \delta K \mapsto (\delta K, 0).$$

By Proposition IV.11, $S_{(x,\xi)} = \ker \sigma_\xi(L)$. By Corollary IV.3, $S_{(x,\xi)} \oplus \{0\} \subset \ker \sigma_\xi(\mathbf{L}_{\mathrm{full}}) \subset \mathscr{U}_x$. By Lemma IV.12, this rank-k subspace survives quotient by $\mathrm{im}\,\sigma_\xi(\mathbf{G})$. Hence $\bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}})$ is non-injective. □

Definition IV.15 (Admissible internal gauge fixing at flat compatible backgrounds)**.** An admissible internal gauge fixing in the background-fixed internal-gauge setting is a first-order operator

$$\mathscr{Q} : T_{\nabla^\oplus}\mathscr{A}_\oplus \to \Gamma(W)$$

whose principal symbol depends only on the normal-connection slot and has zero principal action on the mixed δK slot at backgrounds with $K_0 = 0$.

Theorem IV.16 (No restoration of ellipticity by admissible internal gauge fixing)**.** *Let $\nabla^\oplus \in \mathscr{Z}_{\mathrm{flat}}$ be a flat compatible background with $K_0 = 0$, and let $n \geq 2$. For any admissible internal gauge fixing $\mathscr{Q}$, the augmented principal symbol*

$$\sigma_\xi(\mathbf{L}_{\mathrm{full}} \oplus \mathscr{Q}) : \mathscr{U}_x / \mathrm{im}\,\sigma_\xi(\mathbf{G}) \to \mathscr{V}_x \oplus W_x$$

is non-injective for every $\xi \neq 0$.

Proof. By Theorem IV.14, the reduced kernel contains a nonzero class from $S_{(x,\xi)} \oplus \{0\}$. By admissibility, $\sigma_\xi(\mathscr{Q})$ vanishes on the mixed slot at $K_0 = 0$, so this class remains in the kernel of the reduced augmented symbol. Hence injectivity cannot be restored. □

Corollary IV.17 (No elliptic compatibility sequence at flat compatible backgrounds)**.** *The principal-symbol sequence*

$$\Gamma(\mathscr{P}) \xrightarrow{\mathbf{G}} \Gamma(\mathscr{U}) \xrightarrow{\mathbf{L}_{\text{full}}} \Gamma(\mathscr{V})$$

is not elliptic at flat compatible backgrounds with $K_0 = 0$.

Proof. Ellipticity of the sequence requires injectivity of the reduced symbol of $\mathbf{L}_{\text{full}}$ at every nonzero covector. Theorem IV.14 gives failure of that injectivity. □

Proposition IV.18 (Reduced-symbol coercivity obstruction at flat compatible backgrounds)**.** *At flat compatible backgrounds with $K_0 = 0$ and for $n \geq 2$, there is no constant $c > 0$ such that*

$$\left|\bar{\sigma}_\xi(\mathbf{L}_{\text{full}})u\right| \geq c|u|$$

for all nonzero ξ and all reduced symbol classes $u \in \mathscr{U}_x / \operatorname{im}\sigma_\xi(\mathbf{G})$.

Proof. By Theorem IV.14, for each nonzero ξ there exists a nonzero reduced class u with $\bar{\sigma}_\xi(\mathbf{L}_{\text{full}})u = 0$. Hence any uniform coercive lower bound fails. □

V. FROZEN KERNEL AND DEFORMATION CONSEQUENCES

Throughout this section, work in a frozen flat chart around a flat compatible background; when mixed-slot gauge invisibility is used, impose $K_0 = 0$.

Theorem V.1 (Local Potential Parametrization of Frozen Kernel)**.** *Let U be simply connected and assume the induced connection on $T^*U \otimes N$ is flat and trivialized on U. Then*

$$\ker(D^{\text{Hom}}) \cap \Gamma(\operatorname{Sym}^2 T^*U \otimes N) = \operatorname{Im}\big(\operatorname{Hess} : C^\infty(U,N) \to \Gamma(\operatorname{Sym}^2 T^*U \otimes N)\big).$$

Equivalently, every frozen mixed-kernel variation is locally Hessian-generated.

Proof. Under the flat trivialization hypothesis of Definition IV.1, D^{Hom} coincides with exterior differentiation in the first index. Thus $D^{\text{Hom}}\delta K = 0$ iff, for each fixed index pair (k,α), the 1-form $\sum_i \delta K^\alpha_{ik}\, dx^i$ is closed:

$$\partial_i \delta K^\alpha_{jk} - \partial_j \delta K^\alpha_{ik} = 0.$$

Since U is simply connected, Poincaré lemma gives first potentials V^α_k with

$$\delta K^\alpha_{ik} = \partial_i V^\alpha_k.$$

Symmetry $\delta K^\alpha_{ik} = \delta K^\alpha_{ki}$ yields the integrability condition

$$\partial_i V^\alpha_k = \partial_k V^\alpha_i.$$

A second Poincaré step gives u^α with $V^\alpha_i = \partial_i u^\alpha$, hence

$$\delta K^\alpha_{ik} = \partial_i \partial_k u^\alpha.$$

Conversely, $\operatorname{Hess}(u)$ lies in $\ker D^{\text{Hom}}$ since $\partial_i\partial_j = \partial_j\partial_i$. If the chart is not frozen, connection and curvature terms enter D^{Hom}, so this exactness statement need not hold in this form. □

Proposition V.2 (Local exactness at the frozen mixed stage)**.** *In a frozen flat chart on simply connected U,*

$$0 \to C^\infty(U,N) \xrightarrow{\mathrm{Hess}} \Gamma(\mathrm{Sym}^2 T^*U \otimes N) \xrightarrow{D^{\mathrm{Hom}}} \Gamma(\Lambda^2 T^*U \otimes T^*U \otimes N)$$

is exact at $\Gamma(\mathrm{Sym}^2 T^*U \otimes N)$.

Proof. Exactness at the middle term is exactly Theorem V.1. □

Remark V.3 (Locality of exactness)**.** The exactness statement is local to the flat trivialized regime. Away from this regime, connection and curvature terms in D^{Hom} obstruct this Hessian exactness.

Lemma V.4 (Gauge directions tangent to flat locus)**.** *At $\nabla^\oplus \in \mathscr{Z}_{\mathrm{flat}}$, internal gauge directions lie in* $\ker D\Phi|_{\nabla^\oplus}$.

Proof. By Lemma III.7,

$$\Phi(g^{-1}\cdot\nabla) = g^{-1}\Phi(\nabla)g.$$

Differentiate at $g = \exp(t\phi)$ and $\nabla \in \mathscr{Z}_{\mathrm{flat}}$ with $\Phi(\nabla) = 0$ to obtain

$$D\Phi|_\nabla\big(D(\mathrm{gauge})|_\nabla \phi\big) = 0.$$

□

Definition V.5 (Frozen formal tangent and reduced quotient)**.** At $\nabla^\oplus \in \mathscr{Z}_{\mathrm{flat}}$, define

$$T_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}} := \ker\big(D\Phi|_{\nabla^\oplus}\big) \subset T_{\nabla^\oplus}\mathscr{A}_\oplus.$$

Define the reduced frozen formal tangent quotient by

$$T^{\mathrm{red}}_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}} := T_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}} \Big/ \mathrm{im}\big(D(\mathrm{gauge})|_{\nabla^\oplus}\big).$$

Theorem V.6 (Frozen reduced formal tangent is infinite-dimensional)**.** *Let $n \geq 2$ and let $\nabla^\oplus \in \mathscr{Z}_{\mathrm{flat}}$. On each simply connected frozen flat chart U, the reduced tangent quotient $T^{\mathrm{red}}_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}}$ contains an infinite-dimensional mixed-sector subspace.*

Proof. Let U be simply connected under the flat trivialization hypothesis of Theorem V.1, and impose $K_0 = 0$ on U. For $u \in C^\infty(U,N)$, set $\delta K = \mathrm{Hess}(u)$. By Theorem V.1,

$$D^{\mathrm{Hom}}(\delta K) = 0.$$

Hence $\eta = \iota(0,\delta K) \in \ker(D\Phi|_{\nabla^\oplus})$ by Lemma IV.3. If $\phi \in \Gamma(\mathfrak{so}(N))$ generates an internal gauge direction, then

$$\delta_\phi K = -\phi\cdot K.$$

At $K = 0$, $\delta_\phi K = 0$, so the image of

$$D(\mathrm{gauge})|_{\nabla^\oplus} : \Gamma(\mathfrak{so}(N)) \to T_{\nabla^\oplus}\mathscr{A}_\oplus$$

has zero projection to the mixed slot and cannot kill nonzero Hessian classes. If $\mathrm{Hess}(u) \neq 0$, the class of $\eta = \iota(0,\mathrm{Hess}(u))$ is nonzero in

$$T^{\mathrm{red}}_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}} = T_{\nabla^\oplus}\mathscr{Z}_{\mathrm{flat}} / \mathrm{im}\big(D(\mathrm{gauge})|_{\nabla^\oplus}\big).$$

Since $\{\,\mathrm{Hess}(u) : u \in C^\infty(U,N)\,\}$ is infinite-dimensional, the quotient contains an infinite-dimensional mixed-sector subspace. □

Corollary V.7 (Codimension-one frozen deformation consequence)**.** *For $k = 1$ and $n \geq 2$, the internal gauge group is discrete. At flat compatible backgrounds with $K_0 = 0$, the frozen reduced formal tangent quotient contains an infinite-dimensional mixed-sector subspace generated by Hessian families.*

Proof. For $k = 1$, $\mathrm{SO}(1)$ is trivial at infinitesimal level, so internal gauge contributes no nontrivial quotient directions. Apply Theorem V.6. □

Remark V.8 (Hypersurface-data interpretation)**.** For codimension one, this statement applies to the frozen compatibility sector of hypersurface data with fixed background metric h: the mixed deformation directions are not elliptically controlled at symbol level.

Thus the flat locus has infinite formal codimension in the mixed direction.

VI. RELATION TO THE EINSTEIN CONSTRAINT SYMBOLS

Throughout this section, all statements are principal-symbol only, in flat codimension one with $K_0 = 0$ in a frozen flat chart; no nonlinear equivalence and no diffeomorphism-gauge identification are asserted. The codimension-one case is structurally special: after unit-normal trivialization, $\mathrm{Hom}(TM, N) \cong T^*M$ canonically, which yields the divergence-form contraction on the mixed block.

Definition VI.1 (Codimension-one contraction map)**.** For $k = 1$, the unit-normal identification $N_x \cong \mathbb{R}$ gives a canonical contraction

$$\mathrm{Contr} : \Lambda^2 T^*M \otimes T^*M \to T^*M.$$

In local coordinates, this is

$$(\mathrm{Contr}\, C)_i := h^{jk} C_{ijk}.$$

Theorem VI.2 (Codimension-one divergence reduction at flat compatible background)**.** *Assume $k = 1$ and work at a flat compatible background with $K_0 = 0$ in a frozen flat chart. The statement is principal-symbol only. The linearized mixed defect satisfies*

$$(\delta C)_{ijk} = \partial_i \delta K_{jk} - \partial_j \delta K_{ik}.$$

After contraction,

$$\delta \bar{C}_i := h^{jk} (\delta C)_{ijk} = \partial^j \delta K_{ij} - \partial_i \mathrm{tr}(\delta K).$$

At principal symbol level,

$$\mathrm{Contr} \circ \sigma_\xi (D^{\mathrm{Hom}})(\delta K) = \xi^j \delta K_{ij} - \xi_i \mathrm{tr}(\delta K).$$

Proof. At $k = 1$ the mixed bundle is locally trivialized by a unit normal. Under the flat trivialization hypothesis of Definition IV.1, its principal symbol is $\xi \wedge \cdot$, and in frozen coordinates the full operator agrees with ∂ at leading order, giving the first formula. Contracting with h^{jk} yields the divergence-trace form. Replacing ∂ by ξ gives the symbol identity. □

Corollary VI.3 (Universality of divergence-symbol degeneracy)**.** *At $k = 1$, at a flat compatible background with $K_0 = 0$, fix the local unit-normal identification $N|_U \cong U \times \mathbb{R}$, the metric raising map $h^\sharp : T^*U \to TU$, and*

$$\mathrm{Contr}(C)_i = h^{jk} C_{ijk}.$$

Then, for each nonzero $\xi \in T_x^*U$, the principal-symbol maps

$$\delta K \longmapsto \mathrm{Contr} \circ \sigma_\xi (D^{\mathrm{Hom}})(\delta K), \qquad \delta K \longmapsto \sigma_\xi\big(\mathrm{div}(\delta K) - d\,\mathrm{tr}(\delta K)\big),$$

from $\mathrm{Sym}^2 T_x^*U$ to T_x^*U are equal. Thus the two symbols coincide as linear maps $\mathrm{Sym}^2(T_x^*U) \to T_x^*U$ for every nonzero covector ξ.

Proof. Apply Theorem VI.2 and contract with h^{jk} in a unit-normal trivialization of N. □

A. Hamiltonian-Side Symbol Identification

Definition VI.4 (Scalar Gauss-contracted compatibility operator)**.** In codimension one, define

$$\mathscr{H}_{\mathrm{comp}}(h,K) := \mathrm{scal}_h - |K|_h^2 + (\mathrm{tr}_h K)^2.$$

Remark VI.5. The scalar operator coincides with the hypersurface Hamiltonian constraint: $\mathscr{H}_{\mathrm{GR}} := \mathscr{H}_{\mathrm{comp}}$.

Theorem VI.6 (Codimension-one Hamiltonian symbol comparison at flat compatible background)**.** *Assume $k=1$ and work at a flat compatible background with $K_0 = 0$ in a frozen flat chart. This is a principal-symbol statement only. Then*

$$\sigma_\xi\big(D\mathscr{H}_{\mathrm{comp}}\big) = \sigma_\xi\big(\delta\,\mathrm{scal}_h\big),$$

and this equals the principal symbol of the linearized Hamiltonian constraint:

$$\sigma_\xi\big(\delta\mathscr{H}_{\mathrm{GR}}\big)(\delta h) = |\xi|_h^2\,\mathrm{tr}_h(\delta h) - \xi^i\xi^j\delta h_{ij}.$$

Proof. From Proposition II.2,

$$\mathscr{G} = R^h - (K^* \wedge K).$$

Contracting tangent indices in codimension one gives

$$\mathscr{H}_{\mathrm{comp}} = \mathrm{scal}_h - |K|_h^2 + (\mathrm{tr}_h K)^2.$$

Linearizing at $K=0$, the K-terms are quadratic and contribute zero at first order, so

$$D\mathscr{H}_{\mathrm{comp}}|_{K=0} = \delta\,\mathrm{scal}_h\,.$$

Under the flat trivialization hypothesis of Definition IV.1, with $\delta\Gamma^k_{ij} = \frac{1}{2}(\partial_i\delta h_j{}^k + \partial_j\delta h_i{}^k - \partial^k\delta h_{ij})$, the principal part is

$$(\delta\,\mathrm{scal}_h)_{\mathrm{prin}} = -\partial^i\partial^j\delta h_{ij} + \Delta_h\,\mathrm{tr}_h(\delta h).$$

Replacing ∂ by ξ gives

$$\sigma_\xi(\delta\,\mathrm{scal}_h)(\delta h) = |\xi|_h^2\,\mathrm{tr}_h(\delta h) - \xi^i\xi^j\delta h_{ij}.$$

Since $\mathscr{H}_{\mathrm{GR}} = \mathscr{H}_{\mathrm{comp}}$, this is also the principal symbol of $\delta\mathscr{H}_{\mathrm{GR}}$ at $K=0$. □

Corollary VI.7 (Dual universality at flat codimension-one compatible backgrounds)**.** *At $k=1$, at a flat compatible background with $K_0=0$, at principal-symbol level, the momentum and Hamiltonian symbols coincide with canonical contractions of the mixed and Gauss compatibility blocks:*

$$\textit{mixed Codazzi contraction} \longleftrightarrow \sigma_\xi\big(\mathrm{div}\,\delta K - d\,\mathrm{tr}\,\delta K\big),$$
$$\textit{scalar Gauss contraction} \longleftrightarrow \sigma_\xi\big(\delta\mathscr{H}_{\mathrm{GR}}\big).$$

Proof. The momentum-side identification is Corollary VI.3. The Hamiltonian-side identification is Theorem VI.6. Combining both gives the paired symbol-level universality statement. □

Theorem VI.8 (Unified codimension-one constraint-symbol map on $(\delta h, \delta K)$)**.** *Here δh is an auxiliary variation used solely for symbol-class comparison and does not belong to $T_{\nabla^\oplus}\mathscr{A}_\oplus$, and is not part of the compatibility constraint tangent. Assume $k=1$ and work at a flat compatible background with $K_0=0$ in a frozen flat chart. This is principal-symbol only. Define the joint symbol map*

$$\Sigma_\xi : (\delta h, \delta K) \longmapsto \big(\sigma_\xi\big(\mathrm{Contr}\circ D^{\mathrm{Hom}}\big)(\delta K),\ \sigma_\xi\big(D\mathscr{H}_{\mathrm{comp}}\big)(\delta h)\big) \in T_x^*M \oplus \mathbb{R}.$$

Then

$$\Sigma_\xi(\delta h, \delta K) = \big(\sigma_\xi\big(\mathrm{div}\,\delta K - d\,\mathrm{tr}\,\delta K\big),\ \sigma_\xi\big(\delta\mathscr{H}_{\mathrm{GR}}\big)(\delta h)\big)$$

for every nonzero covector ξ.

Proof. The first component equality is Corollary VI.3. The second component equality is Theorem VI.6. The joint statement is their product on the direct-sum input slot $(\delta h, \delta K)$. □

For a symmetric two-tensor K on (M,h), define

$$\mathscr{M}(K) := \operatorname{div} K - d(\operatorname{tr} K).$$

In the frozen flat codimension-one regime, Hessian families $\delta K = \operatorname{Hess}(u)$ from Theorem V.1 satisfy

$$\sigma_\xi(\mathscr{M})(\delta K)_i = \xi^j \delta K_{ij} - \xi_i \operatorname{tr}(\delta K) = \xi^j(\xi_i \xi_j u) - \xi_i(\xi^j \xi_j u) = 0.$$

Thus Hessian variations lie in the kernel of the momentum principal symbol.

By Theorem VI.6, the scalar Gauss contraction yields the Hamiltonian principal symbol at flat codimension-one compatible backgrounds with $K_0 = 0$.

a. Scope boundary. No nonlinear equivalence, no evolution statement, and no diffeomorphism-gauge identification is asserted.

VII. CONCLUSION

The flatness system of the canonical orthogonal connection exhibits principal-symbol classes coinciding with those of the Einstein constraint equations in codimension one within a purely Riemannian framework with internal normal gauge only. At flat codimension-one compatible backgrounds with $K_0 = 0$, the momentum and Hamiltonian symbol classes coincide with canonical contractions of the mixed and Gauss curvature blocks, and the reduced symbol fails injectivity by a mechanism intrinsic to the compatibility geometry. The non-ellipticity is a structural property of the first-order compatibility-flatness operator and does not depend on any second-order curvature dynamics.

It remains to understand persistence of the mixed kernel under perturbation away from flat backgrounds and whether any global moduli interpretation is available.

Appendix A: Appendix A: Canonical Quadratic Penalties and Mixed-Sector Coupling

a. Constraint-geometry interface. In this appendix, quadratic invariants are recorded as canonical local penalty densities for constraint violation $\Phi(\nabla) \neq 0$. The block-rigidity statement means any canonical quadratic penalty detects the mixed defect sector with fixed coefficient, so mixed coupling cannot be removed within the canonical quadratic class.

1. Quadratic class

Fix

$$\langle A, B \rangle_{\text{can}} := -\operatorname{tr}(AB) \tag{A1}$$

on $\mathfrak{so}(E)$. Work with local densities, locally modulo exact forms:

$$\rho, \rho' \in \Omega^n(M), \qquad \rho \sim \rho' \iff \rho - \rho' = d\beta \text{ for some } \beta \in \Omega^{n-1}(M). \tag{A2}$$

Theorem A.1 (Quadratic Minimality)**.** *In curvature degree two, with the canonical pairing fixed as structural input and locally modulo exact forms, the unique nontrivial quadratic invariant density is*

$$-\operatorname{tr}(\mathscr{F}^{\oplus} \wedge *\mathscr{F}^{\oplus}) \tag{A3}$$

up to an overall constant factor.

Proof. The linear candidate $\mathrm{tr}(\mathscr{F}^{\oplus})$ vanishes identically because $\mathscr{F}^{\oplus}$ is skew-valued. Degree filtration then shows the first nonzero invariant order is quadratic. In the fixed-pairing category, every quadratic local invariant is a scalar multiple of the canonical contraction induced by $-\mathrm{tr}$; locally modulo exact forms this is represented by $-\mathrm{tr}(\mathscr{F}^{\oplus} \wedge *\mathscr{F}^{\oplus})$ (Chern–Weil/invariant-polynomial framework[1,2]). Exact-form ambiguity is quotiented by the stated equivalence relation. □

Remark A.2 (Semisimple caveat)**.** For semisimple non-simple algebras, the space of invariant symmetric bilinear forms can be multidimensional. The theorem fixes the canonical pairing as structural input.

Remark A.3 (Topological caveat)**.** In even dimensions, Chern–Weil representatives (e.g. Pontryagin-type densities) can appear globally as closed forms. Theorem A.1 is a local statement modulo exact forms in the fixed canonical pairing class.

2. Quadratic Irreducibility

Definition A.4 (Quadratic decoupling)**.** A quadratic invariant density in the canonical class is *decoupled* if, after block decomposition

$$\mathscr{F}^{\oplus} = \begin{pmatrix} \mathscr{G} & -\mathscr{C}^{*} \\ \mathscr{C} & \mathscr{S} \end{pmatrix}, \tag{A4}$$

it is pointwise equal to a density depending only on $(\mathscr{G}, \mathscr{S})$ and not on $\mathscr{C}$.

Theorem A.5 (Quadratic Constraint Obstruction)**.** *Any local quadratic gauge-invariant density built from the fixed canonical pairing is a constant multiple of* $-\mathrm{tr}(\mathscr{F}^{\oplus} \wedge *\mathscr{F}^{\oplus})$*. In block form, the mixed sector enters with fixed coefficient* 2*:*

$$\|\mathscr{F}^{\oplus}\|^2 = \|\mathscr{G}\|^2 + 2\|\mathscr{C}\|^2 + \|\mathscr{S}\|^2. \tag{A5}$$

Therefore no nontrivial quadratic invariant suppresses mixed dependence unless $\mathscr{C} \equiv 0$*.*

Proof. Theorem A.1 gives uniqueness of the quadratic invariant class up to scalar. The displayed norm identity is its block expansion under canonical orthogonality. The mixed term is $2\|\mathscr{C}\|^2$ with fixed positive coefficient. Hence suppression of mixed dependence is impossible unless $\mathscr{C} \equiv 0$. □

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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